Gavin Lim

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Education

Cornell University Graduation Date: May 2028

BA in Computer Science, Minor in Robotics

Relevant Coursework: Data Structures, Functional Programming, Foundations of AI, Discrete Structures, Linear Algebra

Interests: Battlebots, Garmin Watches, Hybrid Training, Trophy Trucks, Acoustic Guitar

Work Experience & Research

Cornell Human-Robot Collaboration & Companionship Lab (HRC²) Sep 2026 – Present

Undergraduate Researcher, advised by Prof. Guy Hoffman *Ithaca, New York*

- Developed a one-command autonomous mobile-manipulation pipeline on a Hello Robot Stretch RE1 (ROS 2, Nav2) that navigates, localizes, grasps, and places objects across tables; completed 20 end-to-end runs with no human input.
- Replaced fixed grasps with depth-based perception that fits a table plane, segments cans in the point cloud, and computes drive/reach from the kinematic model; detection error was under 1 cm and grasps spanned a 15×11 cm area.
- Cut AMCL drift along featureless table edges from 16–23 cm to 2–11 cm by remapping with teleop SLAM, then added ArUco alignment with the head RGB-D camera (solvePnP + TF), correcting residual drift of up to 19 cm to ∼1 cm.
- Built a Python pipeline that time-syncs and preprocesses robot camera, joint, and gripper data into datasets for fine-tuning a $\pi_{0.5}$ VLA policy on multi-object pick-and-place, where the lab's combined model reached only 15% success.

Combat Robotics at Cornell Nov 2025 – Present

Mechanical Subteam Member, CV Teleoperation Sub-Lead & New-Member Trainer *Ithaca, New York*

- Built a real-time YOLOv11-Pose computer-vision pipeline converting live-camera body keypoints into throttle, steering, and weapon-trigger commands, mapping the operator's keypoint geometry to control signals for controller-free driving.
- Architected the system's safety layer as CV sublead: a failsafe that instantly cuts all drive and weapon motors whenever body tracking is lost or the operator leaves frame, preventing uncontrolled motion of a live combat robot during testing.
- Engineered *California Roll*, a 1lb antweight vertical drum spinner for NHRL; designed a high-kinetic-energy drum weapon, tuning tip speed and tooth-bite geometry to maximize energy transferred per hit within strict mass and budget limits.
- Designed a 2WD skid-steer chassis in Fusion360 with shock-mounted wheelguards to absorb impacts, pocketed uprights for weight reduction, custom-molded traction wheels, and internal component housing, enabling 180° invertible driving.

MercuryVote May 2026 – Aug 2026

Backend Software Engineer Intern *New York City*

- Migrated the production Next.js platform from Vercel to AWS, standing up a containerized backend on ECS/Fargate behind an Application Load Balancer with ECR, Aurora PostgreSQL, Cognito authentication, and Secrets Manager.
- Implemented append-only, transaction-safe audit logging mirrored to CloudWatch and CloudTrail for financial compliance, and built the end-to-end shareholder voting and ballot system with dynamic, per-question ballots and vote-then-delist.
- Engineered a secret-key partner REST API letting brokerages auto-provision user accounts, and integrated Plaid to link accounts and sync 20+ investment holdings per user into Aurora PostgreSQL with idempotent, duplicate-safe SQL writes.

Acoustic Physics Research, Princeton University Mentorship | May 2024 – Oct 2024

Student Researcher, mentored by Dr. LaNell Williams *Princeton, New Jersey*

- Derived a closed-form stiffness-to-wavelength model ($\lambda = 4\rho\pi^3 r^2 l v^2/k$) from Hooke's Law, Newton's 2nd Law, and SHM; built 3 iterative Python simulations, scaling from 50 arbitrary spring constants to empirical 66,000–80,000 N/m values.
- Digitized published piano-string data with WebPlotDigitizer and benchmarked it against model predictions via NumPy least-squares fits, confirming an inverse stiffness-wavelength trend and attributing divergence to likely string wear.

Projects

GuitarBox | Harvard Summer School, Secondary School Program May 2024 – Aug 2024

- Built *GuitarBox*, an Arduino guitar tuner with a C++ real-time control loop polling 6 pull-up inputs and driving 82–330 Hz tone generation and OLED rendering via Adafruit GFX/SSD1306 over I2C, housed in a Fusion 360 press-fit enclosure.
- Prototyped 5 hardware configurations before a portable, power-bank-powered build; gifted 10 units to Cambridge guitarists.

Skills

Languages: C++, Python, SQL, Java, JavaScript/TypeScript

Robotics & Embedded: ROS 2, Linux (Ubuntu), SSH, Arduino, I2C, PyTorch, NumPy, YOLOv11, $\pi_0/\pi_{0.5}$ VLAs

Systems & Tools: Git, Docker, AWS (ECS, CloudWatch), PostgreSQL, MySQL, REST APIs, React, Next.js